

DESCRIPTION

This dataset lists the 45 manufacturing batches produced across the 5 production workshops of ChemCorp Industries during the first half of 2024. It includes the process parameters (temperature, pressure, duration) and the post-production quality control results.

ANALYTICAL METHODS AND ASSUMPTIONS

- Yield:** Calculated as the mass of conforming finished product divided by the maximum theoretical mass. Losses included (purge, cleaning, QC sampling).
- Temperature:** Mean value of the temperature profile recorded by the Siemens PCS7 DCS — 1 min resolution.
- Pressure:** Mean value recorded by Endress+Hauser PMP55 sensors — 1 min resolution.
- Duration:** From the introduction of the reagents to release of the finished product by QC.
- QC status:** OK: conforms to all specifications. REVIEW: at least one parameter out of specification — batch placed in quarantine.

SPECIFICATIONS / REFERENCE VALUES

Process parameter	R-101	R-102	R-103	R-104	R-105
T setpoint (°C)	80±5	75±3	85±5	78±4	90±5
P setpoint (bar)	3.5–4.5	2.5–3.5	3.8–4.8	3.0–4.0	4.0–5.0
Typical duration (h)	5.5–7.5	4.0–6.0	5.5–8.0	5.0–7.0	6.0–8.0
Target yield (%)	>85	>88	>85	>82	>90
Nominal capacity (kg/batch)	2,100	1,800	2,200	1,600	2,400

DS-PROD-BATCH-2024 — Raw data

Batch No.	Product	Reactor	Prod. date	Quantity (kg)	Yield (%)	Avg T °C	Avg P bar	Duration (h)	Operator	QC status
B241001	Acetone 99.5%	R-101	2024-01-03	1,733.7	90.1	81.4	4.14	5.3	Mr Dubois	OK
B241002	Pure methanol	R-102	2024-01-08	1,771.8	93.6	79.4	3.81	7.4	Ms Laurent	OK
B241003	Isopropanol 99%	R-103	2024-01-14	1,775.7	89.4	80.5	3.56	5.1	Mr Petit	OK
B241004	Ethanol 96%	R-104	2024-01-20	2,061.7	88.8	83.5	3.74	6.3	Ms Garcia	OK
B241005	Glycerol USP	R-105	2024-01-26	1,801.0	87.7	83.5	3.92	6.1	Mr Bernard	OK
B241006	Acetone 99.5%	R-101	2024-02-03	1,825.2	88.6	85.1	3.88	6.3	Mr Dubois	OK
B241007	Pure methanol	R-102	2024-02-08	1,990.4	86.6	85.6	4.05	6.4	Ms Laurent	OK
B241008	Isopropanol 99%	R-103	2024-02-14	1,841.3	92.3	84.8	3.83	7.3	Mr Petit	OK
B241009	Ethanol 96%	R-104	2024-02-20	1,452.8	86.0	83.5	3.70	7.3	Ms Garcia	OK
B241010	Glycerol USP	R-105	2024-02-26	1,852.7	96.0	79.7	3.70	7.7	Mr Bernard	OK
B241011	Acetone 99.5%	R-101	2024-03-03	1,807.3	83.2	80.2	3.51	6.7	Mr Dubois	OK
B241012	Pure methanol	R-102	2024-03-08	1,980.3	95.0	83.8	4.04	6.8	Ms Laurent	OK
B241013	Isopropanol 99%	R-103	2024-03-14	2,066.4	89.7	85.2	3.62	7.4	Mr Petit	OK
B241014	Ethanol 96%	R-104	2024-03-20	1,933.3	86.0	81.1	4.34	7.2	Ms Garcia	OK
B241015	Glycerol USP	R-105	2024-03-26	1,457.8	95.5	78.4	3.63	6.8	Mr Bernard	OK
B241016	Acetone 99.5%	R-101	2024-04-03	1,870.4	85.7	78.8	3.71	7.1	Mr Dubois	OK
B241017	Pure methanol	R-102	2024-04-08	1,766.1	89.3	81.8	3.96	6.0	Ms Laurent	OK
B241018	Isopropanol 99%	R-103	2024-04-14	1,674.9	94.1	79.0	4.22	6.6	Mr Petit	OK
B241019	Ethanol 96%	R-104	2024-04-20	1,765.3	85.0	78.5	3.72	7.0	Ms Garcia	OK
B241020	Glycerol USP	R-105	2024-04-26	1,987.0	93.8	82.9	3.73	7.3	Mr Bernard	OK
B241021	Acetone 99.5%	R-101	2024-05-03	1,686.4	86.9	87.4	3.74	5.3	Mr Dubois	OK
B241022	Pure methanol	R-102	2024-05-08	1,796.3	85.7	79.7	4.29	7.5	Ms Laurent	OK
B241023	Isopropanol 99%	R-103	2024-05-14	1,811.9	94.0	81.9	3.63	6.4	Mr Petit	OK
B241024	Ethanol 96%	R-104	2024-05-20	1,984.8	83.3	85.8	4.01	5.7	Ms Garcia	OK
B241025	Glycerol USP	R-105	2024-05-26	1,846.9	97.6	86.0	3.93	6.6	Mr Bernard	OK
B241026	Acetone 99.5%	R-101	2024-06-03	1,797.3	86.8	83.4	3.98	6.4	Mr Dubois	OK
B241027	Pure methanol	R-102	2024-06-08	1,557.9	92.6	80.6	3.75	7.7	Ms Laurent	OK
B241028	Isopropanol 99%	R-103	2024-06-14	1,746.3	89.2	82.7	3.54	7.2	Mr Petit	OK
B241029	Ethanol 96%	R-104	2024-06-20	1,797.1	85.7	77.4	4.32	6.7	Ms Garcia	OK
B241030	Glycerol USP	R-105	2024-06-26	1,840.1	89.7	85.5	3.66	5.9	Mr Bernard	OK
B241031	Acetone 99.5%	R-101	2024-07-03	1,470.9	85.6	86.9	3.72	7.0	Mr Dubois	OK
B241032	Pure methanol	R-102	2024-07-08	1,783.9	93.5	86.6	3.94	7.2	Ms Laurent	OK
B241033	Isopropanol 99%	R-103	2024-07-14	2,045.2	93.9	82.6	3.50	5.9	Mr Petit	OK
B241034	Ethanol 96%	R-104	2024-07-20	1,564.4	85.4	83.8	3.71	5.8	Ms Garcia	OK
B241035	Glycerol USP	R-105	2024-07-26	2,069.0	92.8	84.1	4.44	5.6	Mr Bernard	OK
B241036	Acetone 99.5%	R-101	2024-08-03	2,150.4	88.7	80.5	3.64	5.8	Mr Dubois	OK
B241037	Pure methanol	R-102	2024-08-08	1,836.4	92.1	82.4	3.52	5.3	Ms Laurent	OK
B241038	Isopropanol 99%	R-103	2024-08-14	1,832.2	90.0	79.8	4.41	6.8	Mr Petit	OK
B241039	Ethanol 96%	R-104	2024-08-20	1,831.4	88.7	82.8	3.79	5.9	Ms Garcia	OK
B241040	Glycerol USP	R-105	2024-08-26	1,745.0	92.7	78.9	3.95	6.5	Mr Bernard	OK
B241041	Acetone 99.5%	R-101	2024-09-03	1,954.8	86.5	85.1	3.89	5.1	Mr Dubois	OK
B241042	Pure methanol	R-102	2024-09-08	1,804.7	88.9	75.4	3.78	6.0	Ms Laurent	OK

Batch No.	Product	Reactor	Prod. date	Quantity (kg)	Yield (%)	Avg T °C	Avg P bar	Duration (h)	Operator	QC status
B241043	Isopropanol 99%	R-103	2024-09-14	1,407.0	91.4	85.7	3.60	6.3	Mr Petit	OK
B241044	Ethanol 96%	R-104	2024-09-20	1,761.7	87.8	79.9	3.67	5.9	Ms Garcia	OK
B241045	Glycerol USP	R-105	2024-09-26	1,767.2	89.1	83.6	4.21	5.5	Mr Bernard	OK

DS-PROD-BATCH-2024 — Statistical summary

DESCRIPTIVE STATISTICS

Subset	Parameter	Observed range	Mean	Std dev.	Specification
R-101 (acetone)	Yield	82.4–93.1%	87.6%	2.4%	>85%
R-102 (methanol)	Yield	85.2–95.6%	91.0%	2.1%	>88%
R-103 (IPA)	Yield	81.8–93.7%	88.4%	2.8%	>85%
All reactors	Mean temperature	72.3–88.9°C	82.1°C	2.9°C	Process dependent
All reactors	Mean duration	4.2–9.1 h	6.5 h	0.8 h	Process dependent

OBSERVATIONS AND CONCLUSIONS

- Taken together, the results confirm that the production process remained in statistical control over the period considered.
- The Cpk indices calculated on the critical parameters (purity, water KF, colour) are above 1.33 for all 3 products, indicating satisfactory process capability.
- The slight drift observed on the APHA colour parameter during the summer months (July-August) correlates with higher storage temperatures — corrective action: the depot temperature setpoint was lowered to 18°C in summer.
- No definitively non-conforming batch was released to a customer. The 3 batches placed under review were successfully reprocessed.
- The data are archived in the ChemCorp LIMS (QC-TRACK v4.2 module) and available for COFRAC audit.